

Sangkyu Lee



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RESEARCH INTEREST

Multimodal Generative Models, Embodied AI, Alignment, Long-term Memory, Continual Learning

EDUCATION

Yonsei University

M.S. in Artificial Intelligence (Advisors: [Youngjae Yu](#), [Jongmin Lee](#))

Mar. 2023 – Feb. 2026

Seoul, Korea

Yonsei University

B.A. in Applied Statistics, B.S. in Computer Science and Engineering (CGPA: 4.06/4.30)

Mar. 2021 – Feb. 2023

Seoul, Korea

Hanyang University

Major in Economics and Finance (Transferred, Includes a 2-year mandatory military service)

Mar. 2017 – Feb. 2021

Seoul, Korea

PROFESSIONAL EXPERIENCE

Research Scientist Intern

Artificial Intelligence and Robotics Institute, KIST (Mentors: [Hwasup Lim](#), [Tackgeun You](#))

Sep. 2025 – Present

Seoul, Korea

Researching the real-time execution of robotic manipulation policies that are aware of accidental failure

Research Assistant

Multimodal Intelligence Research Lab, Yonsei University (PI: [Youngjae Yu](#))

Mar. 2023 – Aug. 2025

Seoul, Korea

Participated as a founding member and took charge of topics related to robotics foundation models

Research Scientist Intern

Language Lab, LG AI Research (Mentors: [Honglak Lee](#), [Hosung Song](#), [Janghoon Han](#))

Sep. 2024 – Aug. 2025

Seoul, Korea

Researched for improving Direct Preference Optimization and autoregressive visual generations

Technical Lead

World Model for Robotics and Vehicle Control Team, maumAI

Jan. 2024 – May 2024

Seongnam, Korea

Led the development project of the world foundation model for robotics and autonomous vehicles

Research Scientist Intern

Foundation Research Team, Naver Cloud (Mentors: [Kang Min Yoo](#), [Sungdong Kim](#))

Jul. 2023 – Jan. 2024

Seongnam, Korea

Participated in the alignment of HyperCLOVA X and researched the self-alignment of large language models

Undergraduate Research Assistant

Big Data Analytics and Learning Lab, Yonsei University (PI: [Noseong Park](#))

Jul. 2022 – Aug. 2022

Seoul, Korea

Researched Neural Operator and Stratonovich SDE for the reverse-time process of diffusion models

PUBLICATIONS

Peer-reviewed Conference Papers

[C5, ECCV 2026] Spanning Tree Autoregressive Visual Generation

Sangkyu Lee, Changho Lee, Janghoon Han, Hosung Song, Tackgeun You, Hwasup Lim, Stanley Jungkyu Choi, Honglak Lee, Youngjae Yu

[C4, NeurIPS 2025] KL Penalty Control via Perturbation for Direct Preference Optimization
Sangkyu Lee, Janghoon Han, Hosung Song, Stanley Jungkyu Choi, Honglak Lee, Youngjae Yu

[C3, ACL 2025] Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues
Youngmin Kim*, Jiwan Chung*, Jisoo Kim, Sunghyun Lee, **Sangkyu Lee**, Junhyeok Kim, Cheoljong Yang, Youngjae Yu

[C2, AACL 2025] MASS: Overcoming Language Bias in Image-Text Matching
Jiwan Chung, Seungwon Lim, **Sangkyu Lee**, Youngjae Yu

[C1, ACL 2024] Aligning Large Language Models by On-Policy Self-Judgment
Sangkyu Lee, Sungdong Kim*, Ashkan Yousefpour, Minjoon Seo, Kang Min Yoo, Youngjae Yu*

Preprints & Technical Reports

[P2, arXiv] Real-Time Execution with Autoregressive Policies
Sangkyu Lee, Seohyeon Park, Tackgeun You, Avi Caciularu, Idan Szpektor, Hwasup Lim, Youngjae Yu

[P1, arXiv] SEAL: Entangled White-box Watermarks on Low-Rank Adaptation
Giyeong Oh, Saejin Kim, Woohyun Cho, **Sangkyu Lee**, Jiwan Chung, Dokyung Song, Youngjae Yu

[T1, arXiv] HyperCLOVA X Technical Report
HyperCLOVA X Team (Contributor; **Sangkyu Lee**)

HONORS AND AWARDS

Partial Scholarship, Yonsei University (Spring 2023)
Academic Honors, Yonsei University (Fall 2021)

ACADEMIC SERVICES

Reviewer, ICLR (2025, 2026), NeurIPS (2026), CoRL (2026)

TEACHING EXPERIENCE

Teaching Assistant, AAI5012: Advanced Video Processing (Fall 2023)
Lecture Instructor, Samsung AI Specialist Course (Jul. 2023)

EXTRACURRICULAR ACTIVITIES

Student Society: Lead of Data Science Team, YBIGTA, Yonsei University (Mar. 2022 – Feb. 2023)
Club Activity: Band Leader (Lead Vocalist), SRGB, Hanyang University (Mar. 2017 – Feb. 2019)

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, R, MATLAB
Frameworks: PyTorch, PyTorch Lightning, JAX
Libraries: Transformers, TRL, Accelerate, DeepSpeed, vLLM, Diffusers, scikit-learn, Optuna
Math Coursework: Real Analysis, Mathematical Statistics, Linear Algebra, Numerical Analysis
Language Proficiency: Korean (Native), English (Fluent), Japanese (Fluent)